Bol: Hinglish-to-English Speaking Coach (PRD)

Oct 26, 2026 · @Amili

1. Overview

Bol is a voice-first web app that helps Hindi-speaking students practise spoken English: the student talks, Bol listens, replies out loud in simple English with Hinglish hints, and corrects the one or two most important mistakes per turn.

Problem. Many Indian students understand English but avoid speaking it because they fear mistakes and have no safe place to practise. Human tutors are costly, and most apps are text-based and do not handle Hinglish speech.

Solution. A tap-to-talk conversation coach built on Gnani AI's speech-to-text and text-to-speech models, which are trained for Indian languages, code-mixed speech and noisy audio.

Context. Bol is the entry for Gnani AI's Great Indian AI Internship Challenge (submissions close Nov 10, 2026, 11:59 PM IST). The entry must be a working application that uses at least one Gnani AI model or API, shown in a public demo video.

2. Goals, non-goals and success criteria

The primary goal is a polished, reliable app that a student can use from a phone to hold a spoken English conversation, receive corrections and see progress, finished and demoed before Nov 10.

Goals

- A smooth voice loop: speak, transcribe, tutor reply, spoken answer, in a few seconds per turn.
- Corrections that are short, kind and easy to understand.
- A clean mobile-first design that looks like a real product.
- A clear 60-second demo that shows a before-and-after improvement.

Non-goals (for this version)

- User accounts, login or payments

- Live streaming transcription (push-to-talk is enough)
- Languages beyond Hindi/Hinglish to English
- The Gnani agent builder, which is out of scope for the challenge
- Storing real personal data or recorded calls

Success criteria

Criterion	Target
Full conversation loop works on a phone browser	10 turns in a row without a crash
Turn latency (stop speaking to reply audio starts)	Under 6 seconds on average
Transcription is usable on Hinglish speech	Tested with 3 or more real speakers
Session summary and progress screen work	Scores saved and charted across sessions
Demo video and public post	Live at least 4 days before the deadline

3. Target users and scenarios

The core user is an Indian college or school student who speaks Hindi or Hinglish daily and wants to speak English with confidence for classes, interviews and jobs.

Persona	Situation	What they need from Bol
Beginner student (Tier 2/3 town)	Understands some English, rarely speaks it	Hinglish hints, slow simple replies, gentle corrections
Intermediate college student	Speaks in short, error-prone sentences	Natural conversation, grammar fixes, a score to track
Final-year job seeker	Nervous about interviews	Interview-topic practice and confidence feedback

Key scenarios

- A student practises daily-life talk for 5 minutes before bed and sees a streak build.
- A student picks the Interview topic the night before a campus placement.

- A student replays a corrected sentence to learn the right pronunciation.

4. User flow

A session is a short loop of turns between a welcome screen and a summary screen.

1. **Welcome.** The student sees the Bol name, a one-line tagline and a Start button.
2. **Setup.** The student picks a level (Beginner or Intermediate) and a topic (Daily life, Interview or Travel).
3. **Greeting.** The tutor speaks a short opening question aloud.
4. **Turn loop.** The student taps the mic, speaks and taps again to stop. The app shows the transcript, then the tutor's spoken reply and a correction card.
5. **Finish.** The student taps Finish at any time, or the session ends after about 8 turns.
6. **Summary.** The app shows scores for fluency, grammar and confidence, the top 3 mistakes and one tip.
7. **Progress.** The student can open a chart of past scores and the daily streak.

Error paths. If the microphone is blocked, audio is empty, or an API call fails, the app shows a friendly message with a Retry button and keeps the conversation so far.

5. Functional requirements

P0 items must work for the demo; P1 items are added if time allows; P2 items are only for later.

ID	Requirement	Priority
F1	Tap-to-talk recording in the browser (record, stop, send)	P0
F2	Speech-to-text through the Gnani speech-to-text API, with the transcript shown on screen	P0
F3	Tutor reply generated by a language model using conversation history, level and topic	P0
F4	Tutor reply spoken through the Gnani text-to-speech API	P0

ID	Requirement	Priority
F5	Correction card showing: You said, Better, Why (max 2 per turn)	P0
F6	Level selector (Beginner, Intermediate) and topic selector (Daily life, Interview, Travel)	P0
F7	End-of-session summary: scores out of 10 for fluency, grammar and confidence, top 3 mistakes, one tip	P0
F8	Session history saved (date, topic, scores) and a Progress screen with a score chart	P1
F9	Daily streak counter	P1
F10	Replay button for the tutor's reply and for corrected sentences	P1
F11	Error handling for denied microphone, empty audio, API failure and slow network, with Retry	P0
F12	Demo mode with pre-written turns that works if an API fails	P1
F13	Live streaming transcription	P2

6. Tutor behaviour and response contract

The tutor is a patient, friendly English coach. These rules go into the language model's system prompt.

- Praise something specific first, then correct.
- Correct only the 1 or 2 most important mistakes per turn.
- Keep each reply to 2 short sentences, then ask one follow-up question on the chosen topic.
- Beginner level: simple English plus a short Hindi or Hinglish hint. Intermediate level: English only, slightly richer vocabulary.
- Never be harsh, never lecture, never ask for personal details.
- Stay on the chosen topic and keep language suitable for students.

The model must return only this JSON on every turn, so the app can speak the reply and draw the correction card:

```
{
  "reply": "Nice try! Tell me what you did yesterday.",
  "corrections": [
    { "original": "I am go market", "better": "I went to the market", "why":
"Use past tense: went" }
  ],
  "encouragement": "Good effort, your sentence was easy to understand."
}
```

For the session summary the model returns: fluency, grammar and confidence scores (0 to 10), the top 3 mistakes, and one tip.

7. UI and design requirements

Bol should feel like a calm, friendly product built for phone screens first, with one clear action on each screen.

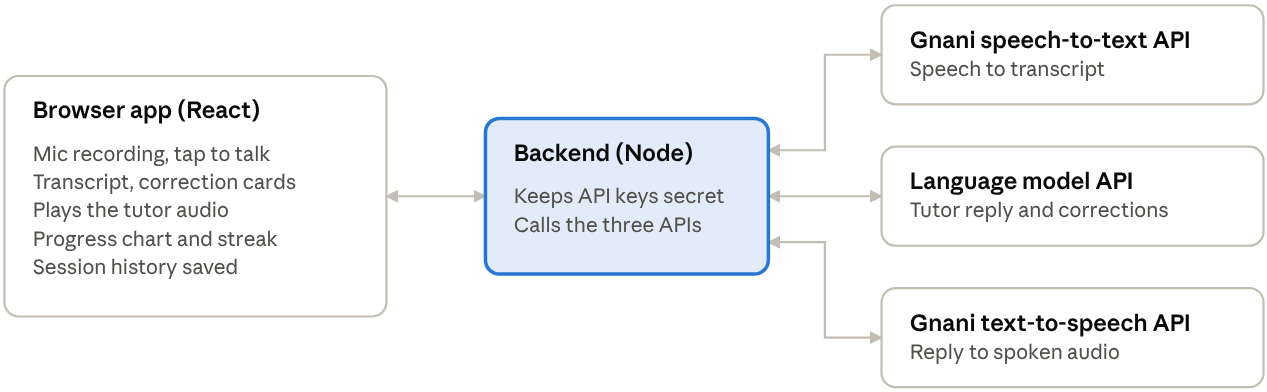
Screen	Key elements
Welcome	App name "Bol", one-line tagline, Start button
Setup	Level and topic selectors as large tappable cards
Conversation	Large centred mic button, live transcript, tutor reply below, sliding correction card, replay button
Summary	Three score rings or bars, top 3 mistakes, one tip, Practise again button
Progress	Line chart of scores over sessions, streak counter

Visual rules

- Calm, friendly colour palette, rounded cards, readable fonts, generous spacing.
- A gentle pulse while recording and a speaking animation while the tutor talks.
- Clear loading states and friendly error text such as "Couldn't hear you, try again".
- Touch targets at least 44 px, and a layout that works from 360 px width upward.
- Corrections read as You said, Better, Why, in plain simple words.

8. Technical architecture

Only the backend holds API keys and calls Gnani and the language model



Each turn runs in order: speech to text, language model, text to speech.

system architecture · browser, backend, 3 APIs

The React app records audio and plays replies, while a small backend holds the API keys and calls Gnani speech-to-text, the language model and Gnani text-to-speech for each turn. The browser never calls these APIs directly.

Layer	Choice	Notes
Frontend	React with Vite and Tailwind CSS	Mobile-first; Recharts for the progress chart
Recording	Browser MediaRecorder API	Push-to-talk: record, stop, send
Backend	Node.js with Express	API keys stored in secrets, never in frontend code
Speech-to-text	Gnani Prisma speech-to-text API	Confirm exact language codes and request format in docs.gnani.ai
Text-to-speech	Gnani Timbre text-to-speech API	Pick a voice that suits Hindi and English
Tutor brain	Hosted language model API	Gnani Evon is not on the API platform, so it is not used here
Storage	Browser local storage (or a small database)	Session history, scores and streak

Layer	Choice	Notes
Build approach	Vibe-coding tool such as Replit Agent	Paste the real Gnani docs into the prompts

Data model

Entity	Fields
Session	id, date, level, topic, scores (fluency, grammar, confidence), top 3 mistakes, tip
Turn	session id, student transcript, tutor reply, corrections (original, better, why)
Progress	streak count, last practice date

9. Non-functional requirements and risks

Non-functional requirements

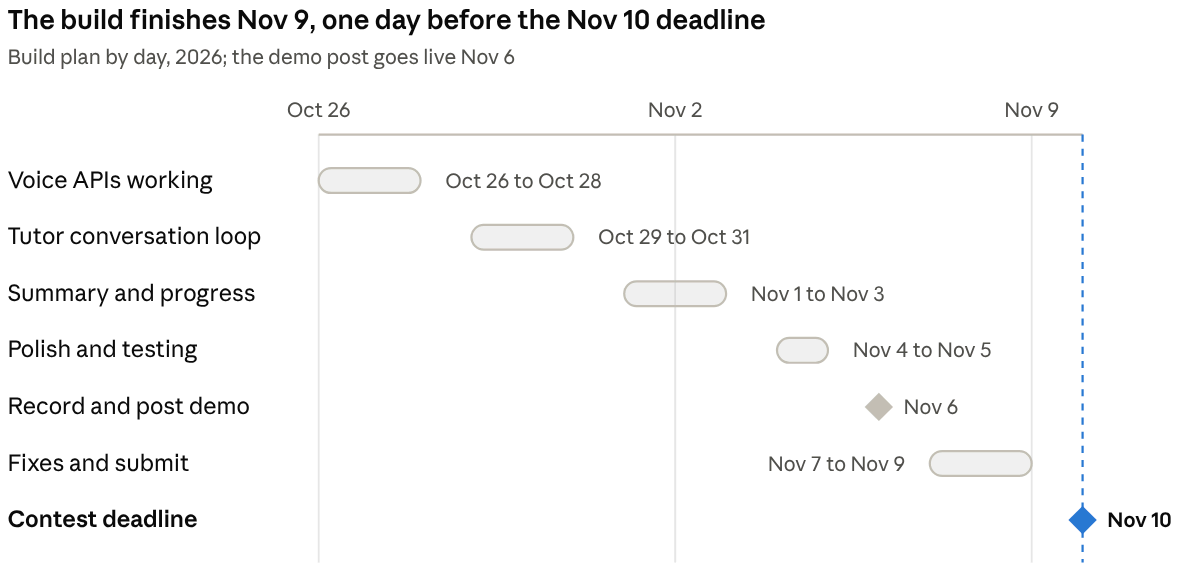
- **Performance:** a turn should feel conversational, with the tutor's audio starting within about 6 seconds of the student stopping.
- **Security:** API keys live only in the backend's secrets; no key appears in frontend code or in chat.
- **Privacy:** use synthetic or made-up data only; do not store real phone numbers, Aadhaar, PAN, account numbers or recorded calls.
- **Compatibility:** works in current mobile Chrome and Safari, and on desktop, over a normal mobile network.
- **Credit use:** stay within the 5,000 programme credits by testing with short sentences and avoiding long test loops.

Risks and mitigations

Risk	Impact	Mitigation
Vibe tool guesses the Gnani API format	Voice calls fail	Paste the real docs and quick-start examples into every voice prompt

Risk	Impact	Mitigation
Hinglish transcription errors	Wrong corrections	Test with real speakers and show the transcript so students can retry
Slow turns	Feels unnatural	Keep replies to 2 sentences; show a loading animation
Credits run out	Cannot demo	Test with short audio; keep a demo mode with pre-written turns
Too many features	Unfinished app	Finish all P0 items before any P1 item
Low engagement on the public post	Weaker shortlist chance	Post the demo around Nov 5 to 6 so reactions can build; no bought or fake engagement
Mic permission or noisy room problems	Session breaks	Clear permission prompt, friendly errors, Retry button

10. Timeline, demo and submission



build timeline · Oct 26 to Nov 10

Building runs from Oct 26 to Nov 9, and the demo post goes live on Nov 6 so it has four days to gather genuine engagement before the deadline.

Demo video (about 60 seconds)

1. 0 to 10 s: state the problem, that many students understand English but are afraid to speak it.
2. 10 to 40 s: a real person speaks a broken sentence, Bol corrects it, and the person repeats it correctly.
3. 40 to 55 s: show the session score and progress screen.
4. 55 to 60 s: close with "Built on Gnani AI's Prisma and Timbre models."

Post the video on LinkedIn, X or both, tag Gnani AI, and add #GreatIndianAIInternshipChallenge and #GnaniAI.

Submission checklist (before Nov 10, 11:59 PM IST)

- [ ] Registered with the correct email and API keys created
- [ ] Project title, problem, users, Gnani models used and each member's contribution filled in
- [ ] Demo video recorded and a written project explanation ready
- [ ] Demo post is live and nominated, one post link per platform
- [ ] Final submission sent at least one day early

Open questions

- [ ] Solo entry or a team of two?
- [ ] Which vibe-coding tool will build it?
- [ ] Which language model API will power the tutor replies?
- [ ] Which Hindi and English voices and language codes does the Gnani API support?